A Comparative Study on Domain-Specific LLMs: Architecture and Performance

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Abstract. The research development on domain-specific large language models (DSLLMs) has expanded rapidly in recent years. Despite this growth, it has also made the landscape more fragmented and difficult to navigate. Although many studies have shown the benefits of adapting large language models to specific domains, the field lacks a unified framework that brings together other researchers' findings and enables meaningful comparisons. This review gathers knowledge from around 60 studies to present a comprehensive analysis of DSLLMs, addressing current knowledge gaps. First, we structure direct performance comparisons across domains, architectures, and metrics through systematic comparison tables, revealing that architectural choice fundamentally determines capability patterns. Although encoder-only models excel at understanding tasks (BERT-Large: 93.2 F1 on SQuAD v1.1), decoder-only models show superior generation capabilities (GPT-3: 87.7% Story Cloze accuracy). Encoder-decoder models provide balanced but sometimes suboptimal performance on highly specialized tasks. Second, we identify a critical gap in temporal knowledge management, demonstrating that existing benchmarks do not evaluate the ability of individual models to track the evolution of knowledge or to recognize outdated information. Third, our analysis reveals that when general-purpose models outperform specialized ones (BERT-Large achieving 96.72 F1 vs. BioBERT's 88.85 F1 on BC5CDR), the benefits are driven by factors such as scale, data diversity, and metric limitations rather than architectural superiority.

Keywords: Domain-Specific Large Language Models (DSLLMs), Large Language Models (LLMs), Transformer Architectures, BERT, GPT, T5, BioBERT, SciBERT, BioGPT, SciFive, mT5, ByT5, Natural Language Processing (NLP)

1 Introduction

The emergence of domain-specific large language models (DSLLMs) represents a natural evolution in natural language processing, driven by the recognition that general-purpose models often fall short when applied to specialized domains with unique vocabularies and contextual requirements. Although the concepts of DSLLMs have roots in the mid-2000s [15], they experienced exponential progress with the increase in transformer-based architectures, such as BERT and GPT [1, 4]. The evolution progressed from basic transfer learning and fine-tuning methods to advanced DSLLMs strategies, eventually leading to models built entirely on DSLLM corpora [16]. The observation of performance gaps motivated this evolution, as when general models were applied to specialized tasks, such as biomedical or scientific tasks, revealing that effective DSLLMs NLP require a deep architectural understanding rather than surface-level adaptation.

1.1 Problem description

The field of DSLLMs has experienced rapid growth in recent years, and this expansion has created significant challenges for researchers and practitioners trying to navigate the increasingly complex landscape of available models and methodologies. Although individual studies have demonstrated the effectiveness of DSLLMs in various fields, the research community would benefit from a comprehensive analysis that combines these findings into a unified framework for understanding and evaluating the performance of DSLLMs. Most of the existing literature focuses on presenting novel models or demonstrating improvements over baseline systems. However, few studies provide the critical analysis necessary to understand the underlying principles that drive successful domain adaptation. This presents an evaluation gap that becomes especially concerning in high-stakes domain applications, where model failures can lead to significant real-world impacts. The need for a comprehensive review that not only reviews existing approaches, but also provides a critical analysis of their strengths, weaknesses, and appropriate applications has become increasingly urgent as the field continues to expand.

1.2 Objectives / Contribution

This comprehensive review represents an attempt to unify, organize, and analyze the diverse landscape of DSLLMs. We summarized findings from around 60 research papers to create a unified framework in this rapidly evolving field. Our work addresses three critical gaps in the current understanding of DSLLMs through systematic analysis and structured comparison.

First, we provide a comprehensive framework that unifies the results of different studies into structured comparison tables, enabling direct performance comparisons between architectures, domains, and evaluation metrics. This enables the identification of hidden patterns in model behavior and provides practitioners with actionable insights for model selection based on task requirements, rather than relying on fragmented individual study results.

Second, we demonstrate that current evaluation frameworks fundamentally do not address the precision of knowledge, which is a critical limitation given that, to the best of our knowledge, existing benchmarks do not directly evaluate a model's ability to handle temporal knowledge or detect outdated information. This gap is concerning in domains where information evolves rapidly, such as medicine, law, and finance, where outdated or misleading knowledge can weaken user trust and real-world deployment.

Third, to the best of our knowledge, no prior work has conducted a systematic comparison at the architectural level. In this study, we provide a comprehensive architectural comparison framework that analyzes how encoder-only, decoder-only, and encoder–decoder architectures produce distinct error patterns and performance results when applied to DSLLM tasks.

1.3 Article sectioning

The remainder of this paper is organized as follows. Section two introduces Large Language Models and their main architectures, such as including encoder-only, decoder-only, and encoder-decoder, along with their DSLLM variants. Section three provides a detailed analysis of BERT and its specialized versions, such as SciBERT and BioBERT. Section four presents the GPT model and its evolution from GPT-1 to GPT-4 and some of its variants. Section five discusses the T5 architecture and its DSLLMs extensions. Section six provides different LLM's metrics and benchmarks. Section seven discusses key findings, open challenges, and future research directions. Finally, Section eight concludes the paper.

2 Introduction to LLMs

Most large Language Models (LLMs) are AI systems based on the transformer architecture, which uses attention mechanisms to capture relationships between all sequence elements at the same time [14] through massive neural networks trained on large datasets to understand and generate human language. These models represent the evolution of the concept of language models in AI, which has made a great impact on the capabilities of AI models. The evolution of simple tasks to more complex systems that can represent text through deep models that are capable of diverse tasks like language translation, summarization, and reasoning capabilities that have been significantly improved through specialized training methods.

2.1 Different LLM models

Encoder-Only Models Encoder-only models are designed primarily for understanding and extracting information from text rather than generating new content. These models excel at tasks that require deep comprehension of the input text, such as classification, recognition of named entities, and answering questions. The most prominent example is BERT, which uses a bidirectional approach to consider both left and right contexts during training [1]. These models work by processing the entire input sequence at once, using self-attention mechanisms to understand relationships between all words in the text. The bidirectional nature allows them to capture richer contextual representations compared to unidirectional approaches [2]. Variants like DistilBERT have been developed to make these models more efficient while maintaining performance, achieving nearly the same results as BERT but with half the parameters [3].

Decoder-Only Models Decoder-only models are primarily designed for text generation tasks, following an autoregressive approach where they predict the next token in a sequence based on previous tokens. These models excel at creative writing, code generation, and conversational tasks. The GPT series represents the most well-known decoder-only architecture, starting with GPT-1's two-stage approach of generative pretraining followed by task-specific fine-tuning[4]. The

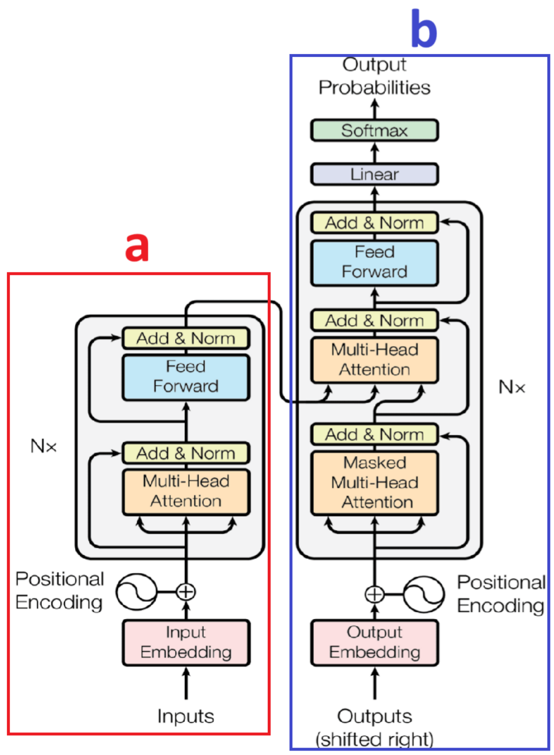

Fig. 1: Illustration of the Transformer architecture: a) The left part in the red box shows the encoder-only structure, which forms the basis of models such as BERT. b)The right part in the blue box depicts the decoder-only structure, characteristic of models like GPT. Together, the full encoder–decoder framework represents the original Transformer model, which underlies models such as T5. [20].

architecture uses masked self-attention to ensure that predictions for each position can only depend on the known outputs at previous positions, maintaining the autoregressive property. This design makes them particularly effective for language generation tasks, as demonstrated by models such as DialoGPT, which extends GPT-2 for the generation of conversational responses[5].

Encoder-Decoder Models Encoder-decoder models combine the strengths of both architectures, using an encoder to understand the input and a decoder to generate the output. This makes them particularly suitable for tasks that require both comprehension and generation, such as machine translation, summarization, and question answering. The encoder processes the input sequence to create contextual representations, while the decoder generates the output sequence using both the encoded representations and the previously generated tokens. Models like T5 [6] exemplify this architecture, treating every NLP task as a text-to-text problem. Multilingual variants such as mT5 extend this approach across multiple languages, achieving SOTA results on multilingual benchmarks. Specialized versions like Sentence-T5 focus on generating high-quality sentence embeddings using encoder-only and encoder-decoder strategies [7].

2.2 Domain Specific LLM

DSLLMs are models trained or fine-tuned on domain-specific corpora (e.g., medical texts, legal documents, scientific papers) to outperform general-purpose models in specialized fields. These models address the gap between general-purpose LLM and specialized domain applications by incorporating domain knowledge and terminology [8]. In the biomedical domain, models like BioBERT and SciBERT have been developed by pre-training on DSLLM corpora such as PubMed articles and scientific papers. BioBERT, adapted from BERT and pre-trained in PubMed and PMC data, significantly outperforms general BERT on biomedical text mining tasks [9]. Similarly, SciBERT, pre-trained in scientific literature, shows notable improvements on scientific NLP tasks [10]. Generative domain-specific models like BioGPT and Galactica represent advances in specialized text generation. BioGPT bridges the gap left by discriminative models in biomedical text generation [11], while Galactica focuses on scientific knowledge, demonstrating the ability to store, combine, and reason about scientific information [12]. The development of DSLLMs often involves careful consideration of specialized datasets, evaluation metrics, and training strategies tailored to the requirements and challenges of the particular domain [13].

2.3 Other Major Models

This paper focuses primarily on BERT, GPT, and T5 architectures and their DSLLM variants. While other significant models, such as LLAMA, CLAUDE, and GEMINI demonstrate impressive general capabilities, the scientific literature on their domain-specific variants is limited, leading to their exclusion from this review.

3 Bert

Table 1: Performance overview of BERT and its variants on standard NLP benchmarks. Metrics include accuracy and F1 scores. The table highlights how specialized DSLLMs such as SciBERT and BioBERT, and other versions of general-purpose models such as Albert and Distilbert, improve the performance, reduce computational cost, or achieve SOTA results at publication time.

Model	**Domain**	**GLUE(AVG)**	**SQuAD v1.1 (F1)**	**SQuAD v2.0 (F1)**	**SWAG (Acc)**	**NER–BC5CDR (F1)**
BERT large [1]	General	82.1	93.2	83.1	86.3	96.7 (BERT-base)
SciBERT [10]	Scientific	–	–	–	–	90.0
BioBERT [9]	Biomedical	–	–	–	–	88.8
DistilBERT [3]	General (compressed)	77.0	86.9	–	–	–
ALBERT base [23]	General (lightweight)	89.4	89.3	80.4	–	–

Architecture Overview BERT (Bidirectional Encoder Representations from Transformers) is built using a multilayer, bidirectional Transformer encoder, closely following the implementation style described by [20]. BERT comes in two primary configurations: BERTBASE, which features 12 layers (transformer blocks), a hidden size of 768, and 12 self-attention heads, totaling 110 million parameters; and BERTLARGE, with 24 layers, a hidden size of 1024, and 16 attention heads, making up 340 million parameters. Unlike models such as GPT, which use attention in the left context only, BERT uses fully bidirectional self-attention, allowing it to capture rich contextual information from both directions within the input [1]. Despite BERT's strong performance on understanding tasks, recent evaluations of academic question-answering reveal its limitations on generation tasks, achieving only a 0.25 BLEU score compared to generative models [53].

Pre-training Tasks The BERT model is pre-trained using two unsupervised prediction objectives: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). Masked LM (MLM): Randomly selected tokens (15% of WordPiece tokens) in each input sequence are masked, and the model is trained to predict the original tokens. To address the discrepancy between pretraining (which sees MASK tokens) and fine-tuning (which does not), the replacement is performed as follows: 80% of the time, tokens are swapped with [MASK]; 10% of the time, with a random token; and 10% of the time, kept unchanged [1]. Next-Sentence Prediction (NSP): BERT is also trained to understand relationships between sentences by predicting whether one sentence actually follows another or if it is a random sentence from the corpus. This binary objective is helpful for later tasks that involve reasoning about sequence relationships, such as question answering and natural language inference [1].

Fine-tuning BERT As discussed in [1], the fine-tuning process is simplified because BERT's Transformer-based self-attention architecture enables it to handle various downstream tasks, both single-text and text-pair scenarios, by simply adjusting the input and output configurations. Traditional approaches [21] [22] for text-pair tasks typically encode each text separately before applying bidirectional cross attention, while [1] unifies these steps by using self-attention on concatenated text pairs, which inherently captures bidirectional cross attention between sentences. Furthermore, BERT adapts to different tasks by plugging in task-specific inputs/outputs while training end-to-end. It reuses its sentence pair structure for various tasks like paraphrasing, entailment, Q&A, and classification, then selects appropriate output representations.

Domain-Specific variants The success of BERT has led to numerous DSLLM variants that leverage the encoder-only framework for specialized applications across different fields. These variants demonstrate the flexibility of BERT's pre-training approach when combined with DSLLMs corpora, training strategies, and can be seen in many domains, such as Scientific and Biomedical.

Scientific Domain variants: SciBERT represents a significant advancement in scientific text processing, utilizing BERT's architecture while being pre-trained on a large corpus of scientific papers [10]. **Biomedical and Clinical variants**: BioBERT exemplifies a successful domain variant by pre-training BERT on PubMed and PMC corpora, creating a model specifically tailored for biomedical text mining [9]. Clinical BERT models have been developed specifically for clinical free text processing, with variants trained on general clinical notes and discharge summaries [46]. CovBERT represents a highly specialized variant for COVID-19 literature classification, demonstrating how BERT can be fine-tuned for emerging DSLLMs needs [54].

4 GPT

Table 2: Performance of GPT-based models Benchmarks cover tasks such as natural language inference, commonsense reasoning, and question answering, with results at the time of publication.

Benchmark	**GPT-1 /[4]**	**GPT-2 (Zero-shot) [17]**	**GPT-3 (Few-shot) [18]**
GLUE (AVG)	72.8	–	–
SuperGLUE (AVG)	–	–	71.8
MNLI-m (ACC)	82.1	–	–
MNLI-mm (ACC)	81.4	–	–
RTE (ACC)	56.0	–	69.0
Story Cloze (ACC)	86.5	–	87.7
LAMBADA (PPL)	–	8.6	1.9
LAMBADA (Acc)	–	63.2	86.4

Architecture Overview The Generative Pre-trained Transformer (GPT) represents a paradigm shift in natural language processing through its decoder-only transformer architecture. Unlike bidirectional models such as BERT, GPT employs a unidirectional approach that generates text in an autoregressive manner, predicting the next token based on the preceding context [4]. This architectural choice enables GPT models to excel in text-generation tasks. The main innovation is centered on a two-phase training approach [4]. First, unsupervised pre-training in which the model learns general language patterns by training on massive amounts of text without specific task labels. Second, supervised fine-tuning, which adapts to particular tasks through training on labeled examples. The model uses a special attention mechanism that only allows each word to ”see” the words that came before it in the sequence. This one-directional approach is crucial because it mimics how humans naturally produce language word by word.

Evolution from GPT-1 to GPT-4 The original GPT model established the foundational framework with a 12-layer decoder-only transformer architecture [4]. GPT-1 demonstrated that generative pre-training could achieve competitive performance across multiple natural language understanding tasks through its unified input representation approach. The model's training involved unsupervised pre-training on the BooksCorpus dataset, followed by task-specific supervised fine-tuning, achieving notable improvements over previous SOTA methods on benchmarks, including the GLUE benchmark.

GPT-2 marked a significant increase in the scale and capability of the model, as described in [17]. With 1.5 billion parameters in its largest configuration, GPT-2 demonstrated remarkable zero-shot task performance across diverse domains without task-specific fine-tuning. The model was trained on WebText, a more diverse and larger dataset than GPT-1, leading to improved generalization capabilities on benchmarks, including LAMBADA, enwik8, and others, as measured by ROUGE, BLEU, and accuracy metrics.

GPT-3 represents a significant jump in the language model capabilities with 175 billion parameters, demonstrating that scale can lead to qualitatively different behaviors [18]. The model's most significant contribution lies in its few-shot learning capabilities, where it can perform new tasks with minimal examples provided in the prompt context. This reduced the need for task-specific fine-tuning in many scenarios, establishing in-context learning as a fundamental capability of sufficiently large language models. GPT-3 performance in various benchmarks.

GPT-4 extends the GPT series with enhanced reasoning capabilities and multimodal functionality, processing both text and images [19]. GPT-4 demonstrates significant improvements in complex reasoning tasks, mathematical problem solving, and code generation. The model exhibits more reliable performance on standardized tests and professional benchmarks compared to previous versions, suggesting architectural refinements and improvements in training methodology beyond the simple scaling of parameters.

Domain-Specific variants The GPT architecture has served as inspiration for many DSLLM variants that leverage the decoder-only framework for specialized applications. DialoGPT represents this approach by pre-training on 147 million Reddit conversational exchanges, creating a model specifically optimized for conversational response generation [5]. This specialization demonstrates how the GPT architecture can be adapted through DSLLMs pre-training datasets to achieve superior performance in targeted applications. BioGPT represents another significant domain variant, specifically pre-trained on large-scale biomedical literature to bridge the gap in biomedical text generation and mining [11]. Unlike discriminative models such as BioBERT, BioGPT maintains the generative capabilities of the GPT architecture while achieving SOTA results in six biomedical NLP tasks, demonstrating the adaptability of the decoder-only transformer approach across scientific domains.

5 T5

Architecture Overview T5 (Text-to-Text Transfer Transformer) represents a new direction in language model design by handling natural language processing tasks as a text-to-text problem [52]. T5 is an encoder-decoder transformer framework, which combines the bidirectional understanding capabilities of encoders

Table 3: Performance of T5-based models and their DSLLM variants on standard NLP benchmarks. Metrics include accuracy and F1 scores. Results highlight SOTA performance at the time of publication.

Model	Domain	GLUE (AVG)	SuperGLUE (AVG)	SQuAD (F1/EM)	XNLI Eng (Acc)	GEM-XSum (BLEU)	PubMed (Acc)
T5 base	General	83.4 [7]	76.2 [52]	92.1/85.4 [6]	–	–	87.1 [51]
mT5 base [6]	Multilingual	83.0 [50]	72.5 [50]	89.6/83.8	75.4	8.4	–
ST5 enc-dec base [7]	Sentence embedding	76.3	–	–	–	–	–
ByT5 base [50]	General	85.3	74.0	–	75.4	11.1	–
SciFive [51]	Biomedical	–	–	–	–	–	86.2

with the autoregressive generation power of decoders. The architecture consists of a stack of encoder layers that process input sequences and decoder layers that generate output sequences, with cross-attention mechanisms. T5 comes in multiple sizes, such as T5-Small, T5-Base, and T5-Large, which range between 60 to 770M parameters, and even more like other variants such as T5-11B (11B parameters), which makes it more flexible across different computational constraints and performance requirements [52].

Pre-training of T5 During pre-training of T5, 15% of the input tokens are replaced with sentinel tokens, and each consecutive corrupted span is replaced by a single sentinel token [52]. The model must then generate the corrupted spans in order, separated by the same sentinel tokens. The pre-training corpus for T5 is the Colossal Clean Crawled Corpus (C4), which is 750GB of text. The C4 dataset undergoes some preprocessing methods to remove duplicates, non-natural language content, and low-quality text, resulting in a high-quality training corpus [52]. This enables T5 to develop robust language understanding and generation capabilities in multiple domains.

Fine-tuning of T5 The T5 framework is different from models that require task-specific architectures or output layers. T5 maintains the same input-output format in all applications and only requires task-specific prompt formatting and output parsing [52]. Fine-tuning also involves training the entire model on task-specific datasets, with the encoder processing task-formatted inputs and the decoder generating structured outputs.

Domain-Specific Variants The architecture of T5 became an inspiration for several significant variants for multilingual and DSLLM applications. For example, mT5, which extends the original framework to 101 languages [6]. mT5 is pre-trained on the mC4 dataset and shows strong cross-lingual transfer capabilities, achieving competitive performance on multilingual benchmarks. Another variant for T5 is UL2, which represents an advanced evolution of the T5 framework, incorporating multiple pre-training objectives, including prefix language modeling, span corruption, and other denoising tasks within a single unified framework [49]. There is also ByT5, which extends the capabilities of T5 to byte-level processing, removes the dependency on subword tokenization, and enables more robust handling of hard languages and noisy text [50]. SciFive also applies the T5 framework to the scientific literature, demonstrating improved performance on biomedical and scientific text processing tasks[51].

6 Comparison

To evaluate a large language model, both quantitative metrics and a practical assessment of output quality are needed. Standard metrics include accuracy, precision, F1 score, BLEU, ROUGE, and perplexity. However, practical evaluation for a high-performing LLM should involve high accuracy for DSLLMs, if it were domain-specific, low hallucination rates, and consistency across different types of queries. Furthermore, benchmark datasets are used to measure accuracy rates, and factual correctness can be assessed through human evaluation or comparison with knowledge-grounded datasets specific to the domain.

6.1 Used Metrics

Table 4: Evaluation Metrics for Large Language Models. This table presents the primary metrics used to assess LLM performance across different task categories.

Metric	Task	Purpose
Perplexity (PPL)	Language Modeling	Measures predictive uncertainty (lower is better).
Cross-Entropy Loss	Training Objective	Measures divergence between predicted and true distributions.
BLEU	Text Generation / Translation	Measures n-gram precision compared to reference translations.
ROUGE	Summarization / Generation	Measures n-gram recall/precision overlap with reference summaries.
METEOR	Generation / Translation	F1-like metric considering synonymy, stemming, and word order.
Accuracy	Classification / QA	Proportion of correct predictions.
F1 Score	Classification / NER / QA	Harmonic mean of precision and recall.
Precision / Recall	Classification / NER	Precision = relevancy of predictions, Recall = coverage of relevant items.
Exact Match (EM)	QA (SQuAD, extractive QA)	Proportion of answers exactly matching the gold span.
Human Evaluation	Text Generation	Fluency, coherence, and factuality were judged by human annotators.

Evaluating DSLLMs involves using a variety of performance measures, each aimed at assessing a distinct aspect of the model's abilities. Choosing the right set of metrics is essential to capture the model's strengths and weaknesses accurately and to enable fair, informative comparisons between different architectures, training strategies, and application domains. Metrics can be categorized into 8 different categories:

Classification Metrics Accuracy measures the proportion of correct predictions and appears consistently in evaluation studies, from BERT variants [23] to cross-lingual models on XNLI benchmarks[24].

Precision, Recall, and F1-Score. F1-Score provides balanced evaluation by combining precision and recall, extensively used in biomedical applications [11], scientific text classification [54]. Precision and Recall provide detailed insights into model effectiveness, especially important for tasks like named entity recognition (NER).

Text Generation Metrics BLEU scores measure n-gram overlap between generated and reference texts [28], widely adopted in machine translation studies [26, 27] and cross-lingual applications [32].

ROUGE scores focus on recall-oriented evaluation for text summarization and generation tasks [29], appearing in GPT-2 assessments [17], question generation studies[55], and domain specialization reviews [8].

METEOR emphasizes semantic matching[30] and appears in question generation benchmarks [55] and inductive reasoning evaluations [33].

BERTScore uses contextual embeddings to assess semantic similarity[31], allowing a more refined evaluation compared to traditional surface-level matching methods[55].

Task-Specific Metrics Exact Match (EM) requires a precise match of responses and is used in question and answer evaluations, including GPT-2 benchmark assessments [17] and scientific text processing studies [13]. Specialized Speech Metrics include phone error rate (PER) and word error rate (WER) for acoustic model evaluation, as demonstrated in Speech-XLNet studies on TIMIT and WSJ datasets [25].

Information Retrieval Metrics MAP (Mean Average Precision) and MRR (Mean Reciprocal Rank) provide ranking-based evaluation metrics, particularly relevant for information retrieval and document ranking tasks [56]. These metrics assess not only correctness but also the ranking quality of the retrieved information.

Computational and Efficiency Metrics The number of parameters indicates model size and computational complexity, essential for comparing model efficiency across different architectures. Training Duration and Hardware Requirements measure computational cost and resource efficiency during model development. FLOPs (Floating Point Operations) and Training Energy quantify computational efficiency and the environmental impact of model training.

Human and Qualitative Evaluation Human evaluation remains essential for assessing subjective aspects of model performance, used in conversational AI studies [5], neural machine translation research [27], and behavioral analysis comparing model decision making to human patterns [48].

6.2 Evaluation benchmarks

Advanced Reasoning and Transfer Benchmarks: Transfer Task Accuracy measures model performance on downstream tasks after pre-training [41], particularly important for sentence embedding evaluation using frameworks like SentEval [7]. Reasoning benchmarks include mathematical reasoning assessments (MMLU, MATH) [43, 44], commonsense QA, and symbolic logic evaluations such as logical entailment and proof-based reasoning datasets, as discussed in comprehensive reasoning surveys [42] and in scientific language models like Galactica

Table 5: Common NLP Benchmarks. These benchmarks evaluate model performance across general, advanced, and DSLLMs tasks.

Benchmark	**Task**	**Purpose**
GLUE	General NLP	Composite score across 9 NLP tasks.
SuperGLUE	Advanced NLP	Harder variant of GLUE with reasoning-heavy tasks.
SQuAD	QA	Standard benchmark for span-based QA tasks.
PubMedQA / BioASQ	Biomedical QA	DSLLMs benchmarks for biomedical question answering.
MMLU	Multi-domain QA	Covers 57 tasks across science, law, medicine, etc.

[12]. BIG-Bench provides diverse reasoning and knowledge tasks for comprehensive evaluation [45].

Domain-Specific Benchmarks: Biomedical: Evaluations of medical NLP applications include PubMedQA and MedMCQA for biomedical question answering [16], as well as the MIMIC-III dataset for clinical text processing [46]. **Scientific**: Includes equation retrieval recall used in mathematical information retrieval tasks such as ARQMath and NTCIR-MathIR, along with specialized scientific QA benchmarks, reflecting the unique requirements of scientific text processing [12].

Comprehensive Benchmarks: GLUE Benchmark provides standardized evaluation in multiple language understanding tasks [34], consistently used for BERT variants [1, 3] and comparative studies [35]. SQuAD benchmarks (versions 1.1 and 2.0) [36, 37] offer a reading comprehension evaluation, used in BERT assessments [1] and ALBERT evaluations [23]. XTREME Benchmark assesses multilingual capabilities [38], particularly relevant for mT5 evaluation [6]. RACE Benchmark evaluates the comprehension of reading from examinations [39], commonly used alongside GLUE and SQuAD for a comprehensive evaluation. MultiNLI measures natural language inference across multiple genres and domains [40]. MLQA (Multilingual Question Answering) evaluates cross-lingual question answering capabilities [47]. Language Modeling Benchmarks include LAMBADA for long-range dependency evaluation and enwik8 for character-level language modeling assessment. MLM (Masked Language Modeling) Accuracy measures pre-training performance on masked token prediction tasks, typically used as a training diagnostic rather than a final evaluation metric.

6.3 Summary

The selection of appropriate evaluation metrics depends on the characteristics of the task, the requirements of the domain, and the intended applications. Classification tasks benefit from F1-score and accuracy metrics, generation tasks require BLEU and ROUGE assessments, while semantic understanding tasks increasingly rely on embedding-based metrics like BERTScore. DSLLM applications often require specialized metrics that reflect the unique characteristics and requirements of the target domain, ensuring a meaningful evaluation within the specific context of the application.

7 Discussion of Results / Future trend

After analyzing the different models, we have discovered, as far as we can tell, that none of the existing benchmarks directly evaluate a model's ability to handle temporal knowledge or detect outdated information. The datasets that the models, such as Bert, GPT, and T5, and their DSLLM variants, are static datasets that cannot capture whether these models would identify when their training knowledge has become outdated. For instance, BioGPT's performance on biomedical tasks may reflect knowledge that was current during training but could now be outdated given the rapid pace of medical research advances.

It is worth noting that general models sometimes outperform DSLLMs in some tasks. Looking at table 1, we notice that BERT-large achieves 96.72 F1 on the BC5CDR biomedical NER task, outperforming SciBERT (90.01) and BioBERT (88.85). However, this performance may reflect different factors rather than a simple superiority of general models. Maybe the larger parameter size of BERT-Large compared to the base-sized variants of SciBERT and BioBERT can provide greater capacity to capture complex patterns. In addition, the difference in the pretraining data and the fine-tuning strategies may also affect. Although sometimes the performance of DSLLMs is lower than that of general models on some specific benchmarks, we cannot forget that the DSLLMs are designed to capture DSLLM patterns that general models might miss. The performance gap observed here may reflect limitations in current evaluation metrics rather than fundamental model capabilities. Our findings suggest that the choice between general and DSLLMs should be driven by the task requirement rather than assuming that the general models are superior.

The observation of the different architectures of the models reveals some differences in capability patterns. Encoder-only models show a consistent strength in understanding tasks; for example, BERT-Large achieves 93.2 F1 on SQuAD v1.1 and 83.1 F1 on SQuAD v2.0, indicating robust reading comprehension capabilities. The gap between the two datasets, where SQuAD v2.0 introduces unanswerable questions, shows that these models face challenges in dealing with uncertainty and deciding when not to provide an answer. Decoder-only models show a different pattern. Looking at table 2, we can see that GPT-3 performs well in generative tasks (87.7% Story Cloze accuracy) but shows more variation on classification benchmarks (e.g., 69.0% RTE accuracy). The improvement from GPT-1 (72.8 GLUE average) to GPT-3 (71.8 SuperGLUE average, despite the greater difficulty of SuperGLUE) demonstrates the advantages of scaling for decoder architectures. The encoder-decoder models demonstrate different performance that combines both understanding and generation capabilities. In table 3, we notice that the T5-base achieves an 82.7 GLUE average and 92.08 F1 on SQuAD. On the generative side, mT5 reaches 8.4 BLEU on GEM-XSum, a notable result considering the complexity of multilingual summarization. Moreover, these models show weaker performance in SuperGLUE (T5: 76.2, mT5: 72.1) compared to their GLUE performance, suggesting that these architectures may struggle with complex reasoning tasks.

DSLLMs variants also vary by architecture. In biomedical tasks, encoder-only models like SciBERT and BioBERT show clear specialization benefits, while encoder-decoder models such as SciFive also benefit (86.28 PubMed accuracy vs T5-base 87.18), indicating that the architecture plays a role in determining how efficiently domain knowledge is integrated and applied. These findings reveal that each architecture has some strengths and limitations. Encoder-only models perform best in classification, decoder-only models at generation, and encoder-decoder models that provide balanced but sometimes less optimal performance on highly specialized tasks.

The field of DSLLMs is rapidly evolving, and several critical research directions are emerging and promising to address fundamental challenges in trustworthiness, reliability, and practical deployment.

Trustworthiness: Researchers should develop a framework that can assess when DSLLMs exceed their boundaries in critical fields such as medicine and law. This includes techniques that automatically flag potential error outputs without requiring constant expert supervision and an automated system that can verify the accuracy of model explanations in specialized contexts.

Reliability: The research community should be committed to improving the performance of DSLLMs in unusual cases that were underrepresented in the training data. This involves developing specialized training techniques and evaluation protocols that systematically test model behavior in rare but critical scenarios and focus on resolving conflicts when multiple DSLLMs are trained on identical corpora.

Practical Deployment: Researchers should develop evaluation metrics that bridge the gap between benchmark performance and real-world deployment challenges. This could be by predicting how current evaluation metric failures would affect real-world deployment.

8 Conclusion

This review has presented a comprehensive synthesis of DSLLMs, drawing on evidence from around 60 studies to organize, compare, and evaluate the field in a unified way. By addressing three persistent gaps, temporal knowledge handling, and architectural comparison, we provide a structured foundation for understanding how specialized models behave across different domains and tasks. We find that the most critical gap lies in temporal knowledge. Current evaluation benchmarks are static and do not measure how models handle evolving information or outdated content, which is a serious limitation in fields such as medicine or law, where the use of outdated knowledge can have real-world impacts. We also clarify that general-purpose models outperform DSLLMs is not due to superiority, but due to factors such as larger scale, pre-training data, and weaknesses in evaluation metrics.

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